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Subject: Software Engineering

Section: 5J

Project Title: Student Club Event Ticketing & Budget Portal

Lab 1: Requirements Engineering & UML Use-Case Modelling

Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall route event budget proposals through a multi-stage approval workflow (Faculty Coordinator → Finance Office → Dean).	High	Pass: Proposal status moves from Faculty Coordinator to Finance Office and then to Dean only after approval at each stage. Fail: Proposal reaches a later stage without previous approval.	Ensures proper authorization and financial control.	Verify that each approval stage is mandatory.
FR-002	Functional	The system shall allow a Club Lead to create and submit an event proposal containing event details and the requested	High	Pass: Club Lead can enter all mandatory event and budget details and successfully submit the proposal. Fail: System allows submission	Provides information required for event and budget evaluation.	Specify all mandatory event and budget fields.

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		budget amount.		with missing mandatory fields.		
FR-003	Functional	The system shall allow attendees to register for an approved student club event and generate a unique QR-code ticket for each confirmed registration.	High	Pass: Successful registration generates one unique QR-code ticket linked to the attendee and event. Fail: No ticket is generated or the same QR code is assigned to multiple registrations.	Enables secure and traceable event ticketing.	Ensure each ticket is uniquely linked to one attendee.
FR-004	Functional	The system shall allow Campus Admin to scan and validate an attendee's QR-code ticket at the event venue.	High	Pass: A valid and unused QR code is accepted and the attendee is marked as checked in. Fail: An invalid or already-used QR code is accepted.	Prevents unauthorized or duplicate entry.	Define the response for invalid or previously scanned tickets.
FR-005	Functional	The system shall notify the Club Lead when an event	Medium	Pass: Club Lead receives a notification whenever the proposal	Keeps the Club Lead informed about	Specify the notification method, such as email or

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		budget proposal is approved, rejected, or requires revision.		status changes to Approved, Rejected, or Revision Required. Fail: No notification is generated after a status change.	proposal progress.	portal notification.
NFR-001	Nonfunctional	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.	High	Pass: Benchmarking confirms QR validation completes in less than 100 ms under simulated peak load. Fail: Validation takes 100 ms or longer.	Ensures fast entry and reduces queues.	Test under realistic peak-load conditions.
NFR-002	Nonfunctional	The system shall enforce role-based access control so that users can access only the event proposal and budget functions authorized	High	Pass: Tests confirm that Club Leads, Faculty Coordinators, Finance Officers, Deans, and Campus Admins can access only their authorized functions.	Protects sensitive event and financial information from unauthorized access.	Verify access permissions for each user role.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
		for their assigned role.		Fail: A user can access or modify a function outside their assigned role.		

Actors and Use Cases:

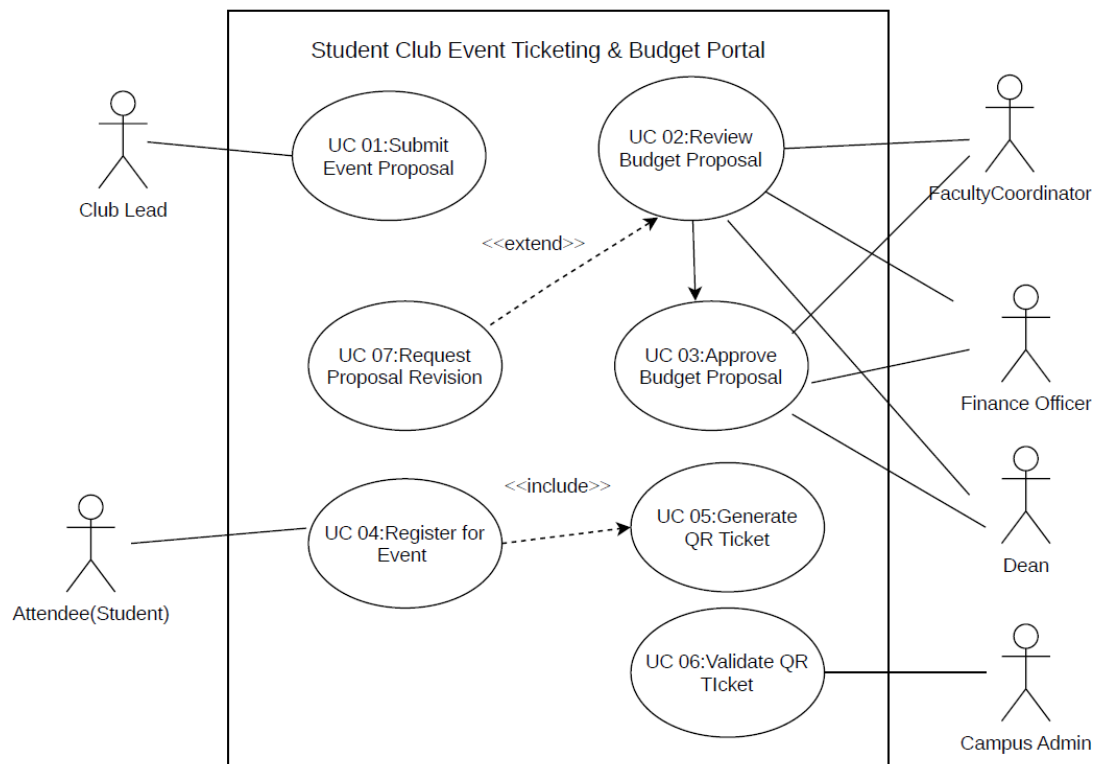
Actors:

1. Club Lead
2. Campus Admin
3. Faculty Coordinator
4. Finance Officer
5. Dean
6. Attendee/Student

Use Cases:

- ❖ Submit Event Proposal
 - Club Lead submits the event and budget proposal.
- ❖ Review/Approve Budget Proposal
 - Faculty Coordinator, Finance Officer and Dean participate in the approval process.
- ❖ Register for Event
 - Attendee registers for an approved event.
- ❖ Generate QR Ticket
 - System generates a ticket after registration/payment/confirmation.
- ❖ Validate QR Ticket
 - Campus Admin scans the QR code at the venue.

UML Use-Case Diagram:



Use-Case Flow Specification:

Use Case Name: Submit and Approve Event Budget Proposal

Primary Actor: Club Lead

Supporting Actors: Faculty Coordinator, Finance Officer, Dean

Preconditions

1. The Club Lead is registered and authenticated in the system.
2. The Club Lead is authorized to submit an event proposal.
3. The Club Lead has the required event details and proposed budget information.
4. The Faculty Coordinator, Finance Officer, and Dean are available to review and approve proposals.

Postconditions

1. The event budget proposal is marked as Approved.
2. The approved budget allocation is recorded in the system.
3. The Club Lead is notified that the proposal has been approved.

4. The proposal status and approval history are stored for future reference.

Main Success Scenario

1. The Club Lead logs into the Student Club Event Ticketing & Budget Portal.
2. The Club Lead selects Submit Event Proposal.
3. The system displays a form for entering event details and the requested budget.
4. The Club Lead enters the required event information and budget amount.
5. The system validates the submitted information.
6. The Club Lead submits the event budget proposal.
7. The system records the proposal and sends it to the Faculty Coordinator for review.
8. The Faculty Coordinator reviews the proposal and approves it.
9. The system forwards the approved proposal to the Finance Office.
10. The Finance Officer reviews the proposed budget and approves it.
11. The system forwards the proposal to the Dean for final approval.
12. The Dean reviews and approves the proposal.
13. The system changes the proposal status to Approved and records the final approval.
14. The system notifies the Club Lead that the event budget proposal has been approved.
15. The use case ends successfully.

Alternate Flow: Proposal Requires Revision

At Step 8, 10, or 12:

1. The Faculty Coordinator, Finance Officer, or Dean identifies an issue with the event proposal or requested budget.
2. The reviewer selects Request Proposal Revision and provides a reason for the revision.
3. The system changes the proposal status to **Revision Required**.
4. The system notifies the Club Lead of the required changes.
5. The Club Lead modifies the proposal and resubmits it.

6. The system restarts the approval process from the appropriate approval stage.
7. The use case continues from the relevant approval step in the Main Success Scenario.